Lab 4 IP, ICMP and TCP Protocol Analysis

This lab is an **individual** lab and can be done at your own time and place. it is important to start the lab before the lab session, since time of 2h might not be enough to complete the lab. During the lab session, you can get instant support/help from the instructor. If you have completed the this lab and do not have questions, it is not required to be present in this lab session.

1. Intended Learning Objectives

At completion of this lab, you will be able to:

1. Explain the IP fragmentation and re-assembly
2. Describe TCP 3-way handshake for connection setup and termination
3. Explain ICMP protocol and how the ping program works
4. Explain how and why the browser uses multiple connections to download the home page and embedded objects.

2. Tasks

2.0 Task 0 Background

Read Lectures on IP and ICMP protocols. Read Lectures on TCP protocol.

Question 1: In IP header, there is a field called “protocol (type)”. What is it used for?

Question 2: How an ICMP message is transported (encapsulation)?

Question 3: Use a figure to show the 3-way handshake to establish a connection in the TCP protocol. What information should be exchanged during this 3-way handshake?

2.1 Task 1 IP fragmentation

This task is about how IP fragmentation and re-assembly work.
(1) Start the WireShark packet capturing
(2) In the command window run the command ping to check if www.home.se is alive. Copy the following command to your command window:

ping -4 -l 3000 www.home.se // for **MS Windows**

where option **-4** is used to force the program to use IPv4; the option **-l 3000** to send the ICMP message of 3000 data bytes.